Лабораторная работа №5

Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

Из проекта Menzhulin-lab4-template была скопирована сеть (рис. 1).

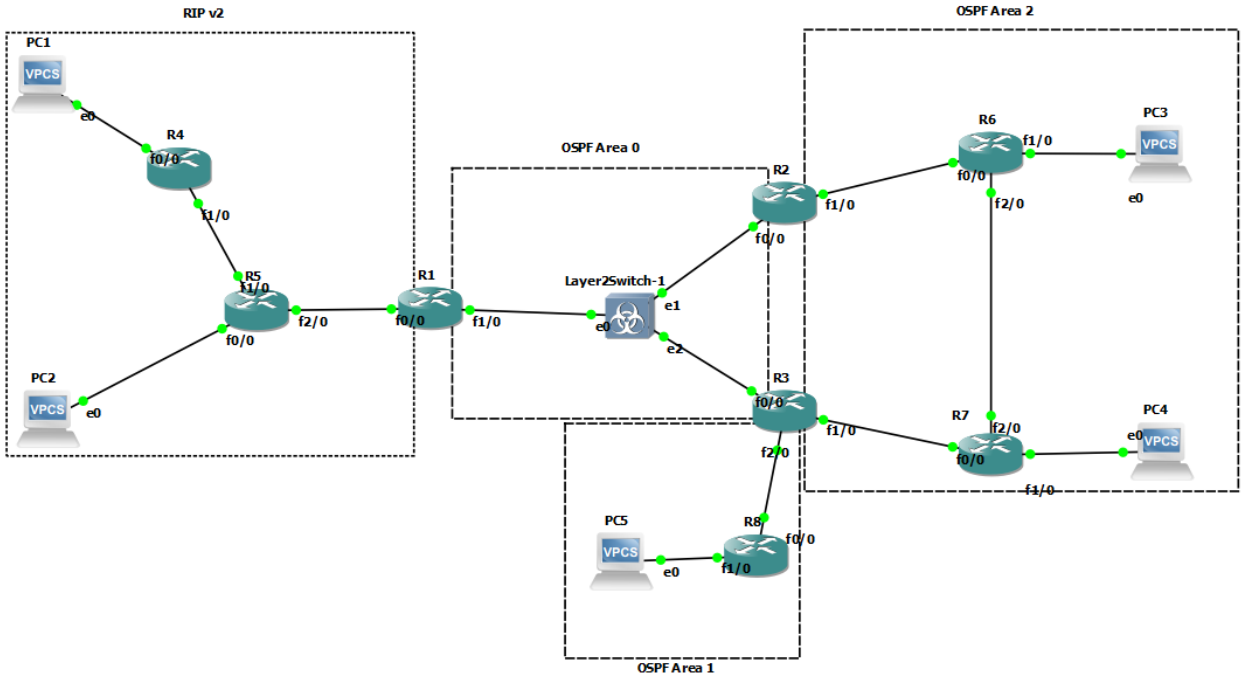

Рис. 1. Топология сети.

Таблица 1. Сети

Сеть	IP-адрес	Маска
PC1 – R4	192.168.1.0	255.255.255.0
PC2 – R5	192.168.2.0	255.255.255.0
R4 – R5	192.168.3.0	255.255.255.0
R5 – R1	192.168.4.0	255.255.255.0
PC5 – R8	192.168.5.0	255.255.255.0

R8 – R3	192.168.6.0	255.255.255.0
PC3 – R6	192.168.7.0	255.255.255.0
PC4 – R7	192.168.8.0	255.255.255.0
R6 – R7	192.168.9.0	255.255.255.0
R6 – R2	192.168.10.0	255.255.255.0
R7 – R3	192.168.11.0	255.255.255.0
R1 – R2 – R3	192.168.12.0	255.255.255.0

Таблица 2. Статические адреса

Устройство	Интерфейс	IP-адрес	Маска
R1	FastEthernet 0/0	192.168.4.1	255.255.255.0
	FastEthernet 1/0	192.168.12.1	255.255.255.0
R2	FastEthernet 0/0	192.168.12.2	255.255.255.0
	FastEthernet 1/0	192.168.10.1	255.255.255.0
R3	FastEthernet 0/0	192.168.12.3	255.255.255.0
	FastEthernet 1/0	192.168.11.1	255.255.255.0
	FastEthernet 2/0	192.168.6.1	255.255.255.0
R4	FastEthernet 0/0	192.168.1.1	255.255.255.0
	FastEthernet 1/0	192.168.3.1	255.255.255.0
R5	FastEthernet 0/0	192.168.2.1	255.255.255.0
	FastEthernet 1/0	192.168.3.2	255.255.255.0
	FastEthernet 2/0	192.168.4.2	255.255.255.0
R6	FastEthernet 0/0	192.168.10.2	255.255.255.0
	FastEthernet 1/0	192.168.7.1	255.255.255.0
	FastEthernet 2/0	192.168.9.1	255.255.255.0
R7	FastEthernet 0/0	192.168.11.2	255.255.255.0
	FastEthernet 1/0	192.168.8.1	255.255.255.0
	FastEthernet 2/0	192.168.9.2	255.255.255.0
R8	FastEthernet 0/0	192.168.6.2	255.255.255.0
	FastEthernet 1/0	192.168.5.1	255.255.255.0
PC1	Ethernet 0/0	192.168.1.10	255.255.255.0
PC2	Ethernet 0/0	192.168.2.10	255.255.255.0
PC3	Ethernet 0/0	192.168.7.10	255.255.255.0
PC4	Ethernet 0/0	192.168.8.10	255.255.255.0
PC5	Ethernet 0/0	192.168.5.10	255.255.255.0

Настройка интерфейсов:

Как пример R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet 0/0
```

```
R1(config-if)#interface FastEthernet 0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface FastEthernet 1/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown
*Mar  1 00:29:50.803: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:29:51.803: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#exit
R1#w
*Mar  1 00:30:01.203: %SYS-5-CONFIG_I: Configured from console by
console
R1#write memory
Building configuration...
[OK]
```

Проверка

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            192.168.4.1     YES manual up
up
FastEthernet1/0            192.168.12.1    YES manual up
up
```

Настройка VPC:

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20792
RHOST:PORT  : 127.0.0.1:20793
MTU         : 1500
```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

На R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.4.0
R1(config-router)#no auto-summary
```

```
R1(config-router)#exit
R1(config)#exit
R1#wri
*Mar  1 01:40:51.279: %SYS-5-CONFIG_I: Configured from console by
console
R1#write memory
Building configuration...
[OK]
```

На R4

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#network 192.168.1.0
R4(config-router)#network 192.168.3.0
R4(config-router)#no auto-summary
R4(config-router)#exit
R4(config)#exit
R4#
*Mar  1 01:34:07.963: %SYS-5-CONFIG_I: Configured from console by
console
R4#write memory
Building configuration...
[OK]
```

На R5

```
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#rou
R5(config)#router r
R5(config)#router rip
R5(config-router)#ver
R5(config-router)#version 2
R5(config-router)#net
R5(config-router)#network 192.168.2.0
R5(config-router)#network 192.168.3.0
R5(config-router)#network 192.168.4.0
R5(config-router)#no aut
R5(config-router)#no auto-summary
R5(config-router)#exit
R5(config)#exit
R5#w
*Mar  1 01:30:03.127: %SYS-5-CONFIG_I: Configured from console by
console
R5#writ
R5#write mem
R5#write memory
Building configuration...
[OK]
```

Проверка записей об маршрутах на маршрутизаторах:

```
R1#show ip route rip
R    192.168.1.0/24 [120/2] via 192.168.4.2, 00:00:15,
FastEthernet0/0
R    192.168.2.0/24 [120/1] via 192.168.4.2, 00:00:15,
FastEthernet0/0
R    192.168.3.0/24 [120/1] via 192.168.4.2, 00:00:15,
FastEthernet0/0

R4#show ip route rip
```

```
R    192.168.4.0/24 [120/1] via 192.168.3.2, 00:00:01,
FastEthernet1/0
R    192.168.2.0/24 [120/1] via 192.168.3.2, 00:00:01,
FastEthernet1/0

R5#show ip route rip
R    192.168.1.0/24 [120/1] via 192.168.3.1, 00:00:05,
FastEthernet1/0
```

Проверка связи между VPC:

```
PC1> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=39.367 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=25.227 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=25.899 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=25.087 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=25.754 ms
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

Настройка area 0:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#exit
R1#wrt
*Mar  1 01:54:03.043: %SYS-5-CONFIG_I: Configured from console by
console
R1#write memory
Building configuration...
[OK]

R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#exit
R2#
*Mar  1 01:50:32.919: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R2#w
*Mar  1 01:50:34.831: %SYS-5-CONFIG_I: Configured from console by
console
R2#write memory
Building configuration...
[OK]

R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
```

```
R3(config-router)#network 192.168.12.0 0.0.0.255 area 0
*Mar  1 01:51:36.011: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar  1 01:51:36.015: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Donet
R3(config-router)#exit
R3(config)#exit
R3#er
*Mar  1 01:51:47.851: %SYS-5-CONFIG_I: Configured from console by
console
R3#write memory
Building configuration...
[OK]
```

Настройка area 2:

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 192.168.10.0 0.0.0.255 area 2
R2(config-router)#exit
R2(config)#exit
R2#
*Mar  1 01:58:54.211: %SYS-5-CONFIG_I: Configured from console by
console
R2#write memory
Building configuration...
[OK]

R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#netw
R3(config-router)#network 192.168.11.0 0.0.0.255 area 2
R3(config-router)#exit
R3(config)#exit
R3#
*Mar  1 02:02:16.335: %SYS-5-CONFIG_I: Configured from console by
console
R3#write memory
Building configuration...
[OK]

R6#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#router ospf 1
R6(config-router)#router-id 6.6.6.6
R6(config-router)#network 192.168.7.0 0.0.0.255 area 2
R6(config-router)#network 192.168.9.0 0.0.0.255 area 2
R6(config-router)#network 192.168.10.0 0.0.0.255 area 2
*Mar  1 01:49:48.291: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R6(config-router)#end
R6#
*Mar  1 01:49:50.995: %SYS-5-CONFIG_I: Configured from console by
console
R6#write memory
Building configuration...
[OK]

R7#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#router ospf 1
```

```
R7(config-router)#router-id
% Incomplete command.

R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 192.168.8.0 0.0.0.255 area 2
R7(config-router)#network 192.168.9.0 0.0.0.255 area 2
R7(config-router)#network 192.168.11.0 0.0.0.255 area 2
R7(config-router)#
*Mar  1 01:50:03.123: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done
R7(config-router)#
*Mar  1 01:50:08.091: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R7(config-router)#end
R7#wr
*Mar  1 01:50:16.031: %SYS-5-CONFIG_I: Configured from console by
console
R7#write memory
Building configuration...
[OK]
```

Настройка area 1:

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 192.168.6.0 0.0.0.255 area 1
R3(config-router)#area 1 stub no-summary
R3(config-router)#end
R3#
*Mar  1 02:18:23.451: %OSPF-5-ADJCHG: Process 1, Nbr 8.8.8.8 on
FastEthernet2/0 from LOADING to FULL, Loading Done
*Mar  1 02:18:23.715: %SYS-5-CONFIG_I: Configured from console by
console
R3#write memory
Building configuration...
[OK]

R8#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#router ospf 1
R8(config-router)#router-id 8.8.8.8
R8(config-router)#network 192.168.5.0 0.0.0.255 area 1
R8(config-router)#network 192.168.6.0 0.0.0.255 area 1
R8(config-router)#area 1 stub
R8(config-router)#end
R8#
*Mar  1 01:56:44.795: %SYS-5-CONFIG_I: Configured from console by
console
R8#write memory
Building configuration...
[OK]
```

Проверки:

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:36    192.168.12.2
FastEthernet1/0
```

```
3.3.3.3           1   FULL/DROTHER    00:00:39    192.168.12.3
FastEthernet1/0
R1#show ip route ospf
O IA 192.168.8.0/24 [110/3] via 192.168.12.3, 00:15:36, FastEthernet1/0
O IA 192.168.9.0/24 [110/3] via 192.168.12.3, 00:15:36, FastEthernet1/0
                    [110/3] via 192.168.12.2, 00:15:36, FastEthernet1/0
O IA 192.168.10.0/24 [110/2] via 192.168.12.2, 00:15:36, FastEthernet1/0
O IA 192.168.11.0/24 [110/2] via 192.168.12.3, 00:15:41, FastEthernet1/0
O IA 192.168.5.0/24 [110/3] via 192.168.12.3, 00:04:03, FastEthernet1/0
O IA 192.168.6.0/24 [110/2] via 192.168.12.3, 00:05:03, FastEthernet1/0
O IA 192.168.7.0/24 [110/3] via 192.168.12.2, 00:15:36, FastEthernet1/0

R2#show ip route ospf
O    192.168.8.0/24 [110/3] via 192.168.10.2, 00:06:09, FastEthernet1/0
O    192.168.9.0/24 [110/2] via 192.168.10.2, 00:06:09, FastEthernet1/0
O    192.168.11.0/24 [110/3] via 192.168.10.2, 00:06:09, FastEthernet1/0
O IA 192.168.5.0/24 [110/3] via 192.168.12.3, 00:04:34, FastEthernet0/0
O IA 192.168.6.0/24 [110/2] via 192.168.12.3, 00:05:35, FastEthernet0/0
O    192.168.7.0/24 [110/2] via 192.168.10.2, 00:06:09, FastEthernet1/0
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:39    192.168.12.1
FastEthernet0/0
3.3.3.3           1   FULL/DROTHER    00:00:32    192.168.12.3
FastEthernet0/0
6.6.6.6           1   FULL/BDR        00:00:39    192.168.10.2
FastEthernet1/0

R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:37    192.168.12.1
FastEthernet0/0
2.2.2.2           1   FULL/BDR        00:00:36    192.168.12.2
FastEthernet0/0
8.8.8.8           1   FULL/DR         00:00:37    192.168.6.2
FastEthernet2/0
7.7.7.7           1   FULL/BDR        00:00:38    192.168.11.2
FastEthernet1/0
R3#show ip route ospf
O    192.168.8.0/24 [110/2] via 192.168.11.2, 00:05:06, FastEthernet1/0
O    192.168.9.0/24 [110/2] via 192.168.11.2, 00:05:06, FastEthernet1/0
O    192.168.10.0/24 [110/3] via 192.168.11.2, 00:05:06, FastEthernet1/0
O    192.168.5.0/24 [110/2] via 192.168.6.2, 00:05:01, FastEthernet2/0
O    192.168.7.0/24 [110/3] via 192.168.11.2, 00:05:06, FastEthernet1/0

R6#show ip route ospf
O IA 192.168.12.0/24 [110/2] via 192.168.10.1, 00:16:43, FastEthernet0/0
O    192.168.8.0/24 [110/2] via 192.168.9.2, 00:16:43, FastEthernet2/0
O    192.168.11.0/24 [110/2] via 192.168.9.2, 00:16:43, FastEthernet2/0
O IA 192.168.5.0/24 [110/4] via 192.168.10.1, 00:05:15, FastEthernet0/0
                    [110/4] via 192.168.9.2, 00:05:15, FastEthernet2/0
O IA 192.168.6.0/24 [110/3] via 192.168.10.1, 00:06:15, FastEthernet0/0
                    [110/3] via 192.168.9.2, 00:06:15, FastEthernet2/0
R6#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:33    192.168.10.1
FastEthernet0/0
7.7.7.7           1   FULL/BDR        00:00:35    192.168.9.2
FastEthernet2/0

R7#show ip route ospf
```

```
O IA 192.168.12.0/24 [110/2] via 192.168.11.1, 00:17:32, FastEthernet0/0
O    192.168.10.0/24 [110/2] via 192.168.9.1, 00:17:32, FastEthernet2/0
O IA 192.168.5.0/24 [110/3] via 192.168.11.1, 00:06:00, FastEthernet0/0
O IA 192.168.6.0/24 [110/2] via 192.168.11.1, 00:07:00, FastEthernet0/0
O    192.168.7.0/24 [110/2] via 192.168.9.1, 00:17:32, FastEthernet2/0
R7#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/DR         00:00:39    192.168.11.1
FastEthernet0/0
6.6.6.6           1   FULL/DR         00:00:34    192.168.9.1
FastEthernet2/0

R8#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:37    192.168.6.1
FastEthernet0/0
R8#show ip route ospf
O*IA 0.0.0.0/0 [110/2] via 192.168.6.1, 00:06:20, FastEthernet0/0
```

Проверка связи между ПК:

С ПК3 пинг ПК4 и ПК5:

```
PC3> ping 192.168.8.10

84 bytes from 192.168.8.10 icmp_seq=1 ttl=62 time=39.073 ms
84 bytes from 192.168.8.10 icmp_seq=2 ttl=62 time=26.571 ms
84 bytes from 192.168.8.10 icmp_seq=3 ttl=62 time=25.913 ms
84 bytes from 192.168.8.10 icmp_seq=4 ttl=62 time=25.697 ms
84 bytes from 192.168.8.10 icmp_seq=5 ttl=62 time=25.402 ms

PC3> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=60 time=42.466 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=60 time=55.135 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=55.144 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=45.398 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=46.092 ms
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

Редистрибюция между протоколами делается на R1, т.к. он является граничным между этими двумя протоколами.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#redistribute ospf 1 metric 2
R1(config-router)#exit
R1(config)#router ospf 1
R1(config-router)#redistribute rip subnets metric 30
R1(config-router)#exi
R1(config)#exit
R1#
```

```
*Mar  1 02:35:57.995: %SYS-5-CONFIG_I: Configured from console by
console
R1#write memory
Building configuration...
[OK]
```

Проверка

На R6:

R6#show ip route

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.12.0/24 [110/2] via 192.168.10.1, 00:26:42, FastEthernet0/0
O    192.168.8.0/24 [110/2] via 192.168.9.2, 00:26:42, FastEthernet2/0
C    192.168.9.0/24 is directly connected, FastEthernet2/0
C    192.168.10.0/24 is directly connected, FastEthernet0/0
O    192.168.11.0/24 [110/2] via 192.168.9.2, 00:26:42, FastEthernet2/0
O E2 192.168.4.0/24 [110/30] via 192.168.10.1, 00:03:18, FastEthernet0/0
O IA 192.168.5.0/24 [110/4] via 192.168.10.1, 00:15:14, FastEthernet0/0
                    [110/4] via 192.168.9.2, 00:15:15, FastEthernet2/0
O IA 192.168.6.0/24 [110/3] via 192.168.10.1, 00:16:15, FastEthernet0/0
                    [110/3] via 192.168.9.2, 00:16:15, FastEthernet2/0
C    192.168.7.0/24 is directly connected, FastEthernet1/0
O E2 192.168.1.0/24 [110/30] via 192.168.10.1, 00:03:19, FastEthernet0/0
O E2 192.168.2.0/24 [110/30] via 192.168.10.1, 00:03:19, FastEthernet0/0
O E2 192.168.3.0/24 [110/30] via 192.168.10.1, 00:03:22, FastEthernet0/0
```

На R4:

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.12.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.8.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.9.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.10.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.11.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
```

```
R    192.168.4.0/24 [120/1] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.5.0/24 [120/3] via 192.168.3.2, 00:00:25, FastEthernet1/0
R    192.168.6.0/24 [120/3] via 192.168.3.2, 00:00:01, FastEthernet1/0
R    192.168.7.0/24 [120/3] via 192.168.3.2, 00:00:01, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 192.168.3.2, 00:00:01, FastEthernet1/0
C    192.168.3.0/24 is directly connected, FastEthernet1/0
```

Как видно в RIP домене и OSPF area 2 появились новые маршруты.

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

```
PC1> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=50.115 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=25.397 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=35.122 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=35.087 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=24.842 ms

PC1> ping 192.168.7.10

84 bytes from 192.168.7.10 icmp_seq=1 ttl=59 time=85.959 ms
84 bytes from 192.168.7.10 icmp_seq=2 ttl=59 time=65.080 ms
84 bytes from 192.168.7.10 icmp_seq=3 ttl=59 time=55.578 ms
84 bytes from 192.168.7.10 icmp_seq=4 ttl=59 time=65.640 ms
84 bytes from 192.168.7.10 icmp_seq=5 ttl=59 time=55.432 ms

PC1> ping 192.168.8.10

84 bytes from 192.168.8.10 icmp_seq=1 ttl=59 time=89.198 ms
84 bytes from 192.168.8.10 icmp_seq=2 ttl=59 time=65.248 ms
84 bytes from 192.168.8.10 icmp_seq=3 ttl=59 time=55.826 ms
84 bytes from 192.168.8.10 icmp_seq=4 ttl=59 time=55.884 ms
84 bytes from 192.168.8.10 icmp_seq=5 ttl=59 time=55.845 ms

PC1> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=59 time=90.871 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=59 time=66.871 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=59 time=65.224 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=59 time=65.920 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=59 time=65.306 ms

PC5> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=59 time=53.575 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=59 time=55.740 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=59 time=56.006 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=59 time=56.095 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=59 time=55.970 ms

PC5> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=44.657 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=45.200 ms
```

```
84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=45.925 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=45.757 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=45.315 ms

PC5> ping 192.168.7.10

84 bytes from 192.168.7.10 icmp_seq=1 ttl=60 time=48.559 ms
84 bytes from 192.168.7.10 icmp_seq=2 ttl=60 time=45.931 ms
84 bytes from 192.168.7.10 icmp_seq=3 ttl=60 time=45.316 ms
84 bytes from 192.168.7.10 icmp_seq=4 ttl=60 time=56.155 ms
84 bytes from 192.168.7.10 icmp_seq=5 ttl=60 time=45.818 ms

PC5> ping 192.168.8.10

84 bytes from 192.168.8.10 icmp_seq=1 ttl=61 time=32.465 ms
84 bytes from 192.168.8.10 icmp_seq=2 ttl=61 time=35.914 ms
84 bytes from 192.168.8.10 icmp_seq=3 ttl=61 time=35.940 ms
84 bytes from 192.168.8.10 icmp_seq=4 ttl=61 time=36.166 ms
84 bytes from 192.168.8.10 icmp_seq=5 ttl=61 time=34.819 ms
```

Как видно, сообщения доходят от каждого VPC до другого.

6) Перехватить в wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

rip

No.	Time	Source	Destination	Protocol	Lengtl	Info
3	7.373793	192.168.4.2	224.0.0.9	RIPv2	106	Response
9	24.964383	192.168.4.1	224.0.0.9	RIPv2	206	Response
12	34.654703	192.168.4.2	224.0.0.9	RIPv2	106	Response
16	51.349435	192.168.4.1	224.0.0.9	RIPv2	206	Response
19	61.240940	192.168.4.2	224.0.0.9	RIPv2	106	Response
26	79.705737	192.168.4.1	224.0.0.9	RIPv2	206	Response
29	89.778799	192.168.4.2	224.0.0.9	RIPv2	106	Response
33	105.849929	192.168.4.1	224.0.0.9	RIPv2	206	Response
37	119.373116	192.168.4.2	224.0.0.9	RIPv2	106	Response
41	132.678667	192.168.4.1	224.0.0.9	RIPv2	206	Response
51	149.248671	192.168.4.2	224.0.0.9	RIPv2	106	Response
68	159.628469	192.168.4.1	224.0.0.9	RIPv2	206	Response
74	177.576838	192.168.4.2	224.0.0.9	RIPv2	106	Response
79	188.640274	192.168.4.1	224.0.0.9	RIPv2	206	Response
83	204.576141	192.168.4.2	224.0.0.9	RIPv2	106	Response
88	218.537720	192.168.4.1	224.0.0.9	RIPv2	206	Response
91	234.723576	192.168.4.2	224.0.0.9	RIPv2	106	Response

Рис. 2. Захваченные пакеты протокола RIP.

Пакеты RIP:

Используют групповую рассылку multicast), на что указывает адрес назначения (224.0.0.9).

```
Frame 3: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits)
on interface -, id 0
```

```
Ethernet II, Src: cc:05:63:7b:00:20 (cc:05:63:7b:00:20), Dst: IPv4mcast_09
(01:00:5e:00:00:09)
    Destination: IPv4mcast_09 (01:00:5e:00:00:09) – многоадресное вещание
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: cc:05:63:7b:00:20 (cc:05:63:7b:00:20)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 2]
Internet Protocol Version 4, Src: 192.168.4.2, Dst: 224.0.0.9
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
        1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable
Transport (0)
    Total Length: 92
    Identification: 0x0000 (0)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 2 – У пакета есть время жизни
        [Expert Info (Note/Sequence): "Time To Live" != 1 for a packet sent
to the Local Network Control Block (see RFC 3171)]
            ["Time To Live" != 1 for a packet sent to the Local Network
Control Block (see RFC 3171)]
            [Severity level: Note]
            [Group: Sequence]
    Protocol: UDP (17)
    Header Checksum: 0x131e [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.4.2
    Destination Address: 224.0.0.9
    [Stream index: 0]
User Datagram Protocol, Src Port: 520, Dst Port: 520
    Source Port: 520
    Destination Port: 520
    Length: 72
    Checksum: 0x0f91 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
    [Timestamps]
    UDP payload (64 bytes)
Routing Information Protocol
    Command: Response (2) – Тип пакета
    Version: RIPv2 (2) - версия
    IP Address: 192.168.1.0, Metric: 2 – сеть 1
        Address Family: IP (2)
        Route Tag: 0
        IP Address: 192.168.1.0
        Netmask: 255.255.255.0
        Next Hop: 0.0.0.0
        Metric: 2
    IP Address: 192.168.2.0, Metric: 1 – сеть 2
```

```
        Address Family: IP (2)
        Route Tag: 0
        IP Address: 192.168.2.0
        Netmask: 255.255.255.0
        Next Hop: 0.0.0.0
        Metric: 1
    IP Address: 192.168.3.0, Metric: 1
```
– сеть 3

```
        Address Family: IP (2)
        Route Tag: 0
        IP Address: 192.168.3.0
        Netmask: 255.255.255.0
        Next Hop: 0.0.0.0
        Metric: 1
```

ospf

No.	Time	Source	Destination	Protocol	Lengtl	Info
102	93.555801	192.168.12.2	224.0.0.5	OSPF	98	Hello Packet
104	94.112142	192.168.12.1	224.0.0.5	OSPF	98	Hello Packet
108	96.599819	192.168.12.3	224.0.0.5	OSPF	98	Hello Packet
114	103.556690	192.168.12.2	224.0.0.5	OSPF	98	Hello Packet
116	104.110240	192.168.12.1	224.0.0.5	OSPF	98	Hello Packet
119	106.598439	192.168.12.3	224.0.0.5	OSPF	98	Hello Packet
126	113.555618	192.168.12.2	224.0.0.5	OSPF	98	Hello Packet
128	114.109627	192.168.12.1	224.0.0.5	OSPF	98	Hello Packet
131	116.598433	192.168.12.3	224.0.0.5	OSPF	98	Hello Packet
136	123.554984	192.168.12.2	224.0.0.5	OSPF	98	Hello Packet
138	124.108831	192.168.12.1	224.0.0.5	OSPF	98	Hello Packet
139	125.808235	192.168.12.2	224.0.0.5	OSPF	90	LS Update
140	125.848523	192.168.12.2	224.0.0.5	OSPF	90	LS Update
143	126.607063	192.168.12.3	224.0.0.5	OSPF	98	Hello Packet
145	128.293409	192.168.12.1	224.0.0.5	OSPF	98	LS Acknowledge
146	128.327196	192.168.12.3	224.0.0.6	OSPF	98	LS Acknowledge

Рис. 3. Захваченные пакеты протокола OSPF.

Пакеты OSPF:

Используют групповую рассылку multicast), на что указывает адрес назначения (224.0.0.5). Есть несколько видов

Пакет hello:

```
Frame 138: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits)
on interface -, id 0
Ethernet II, Src: cc:01:63:03:00:10 (cc:01:63:03:00:10), Dst: IPv4mcast_05
(01:00:5e:00:00:05)
    Destination: IPv4mcast_05 (01:00:5e:00:00:05)
```
– групповое вещание

```
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: cc:01:63:03:00:10 (cc:01:63:03:00:10)
    Type: IPv4 (0x0800)
    [Stream index: 3]
Internet Protocol Version 4, Src: 192.168.12.1, Dst: 224.0.0.5
    0100 .... = Version: 4
```

```
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
        1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable
Transport (0)
    Total Length: 84
    Identification: 0x018f (399)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 1
    Protocol: OSPF IGP (89)
    Header Checksum: 0x0a54 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.12.1
    Destination Address: 224.0.0.5
    [Stream index: 1]
Open Shortest Path First
    OSPF Header
        Version: 2
        Message Type: Hello Packet (1)
```
- тип

```
        Packet Length: 52
        Source OSPF Router: 1.1.1.1
```
– источник, как заданный в настройках

```
        Area ID: 0.0.0.0 (Backbone)
        Checksum: 0x4736 [correct]
        Instance ID: Base IPv4 Unicast Instance (0)
        Auth Type: Null (0)
        Auth Data (none): 0000000000000000
    OSPF Hello Packet
        Network Mask: 255.255.255.0
        Hello Interval [sec]: 10
        Options: 0x12, (L) LLS Data block, (E) External Routing
            0... .... = DN: Not set
            .0.. .... = (O) Opaque: Not set
            ..0. .... = (DC) Demand Circuits: Not supported
            ...1 .... = (L) LLS Data block: Present
            .... 0... = (N) NSSA: Not supported
            .... .0.. = (MC) Multicast: Not capable
            .... ..1. = (E) External Routing: Capable
            .... ...0 = (MT) Multi-Topology Routing: No
        Router Priority: 1
        Router Dead Interval [sec]: 40
        Designated Router: 192.168.12.1
        Backup Designated Router: 192.168.12.2
        Active Neighbor: 2.2.2.2
```
– соседи

```
        Active Neighbor: 3.3.3.3
    OSPF LLS Data Block
        Checksum: 0xfff6
        LLS Data Length: 12 bytes
        Extended options TLV
            TLV Type: 1
            TLV Length: 4
            Options: 0x00000001, (LR) LSDB Resynchronization
                .... .... .... .... .... .... .... ..0. = (RS) Restart
Signal: Not set
                .... .... .... .... .... .... .... ...1 = (LR) LSDB
Resynchronization: Set
```

LS Update: предлагает другим маршрутизаторам какие-то данные

```
Frame 139: Packet, 90 bytes on wire (720 bits), 90 bytes captured
(720 bits) on interface -, id 0
Ethernet II, Src: cc:02:63:21:00:00 (cc:02:63:21:00:00), Dst:
IPv4mcast_05 (01:00:5e:00:00:05)
    Destination: IPv4mcast_05 (01:00:5e:00:00:05)
        .... ..0. .... .... .... .... = LG bit: Globally unique
address (factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: cc:02:63:21:00:00 (cc:02:63:21:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique
address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address
(unicast)
    Type: IPv4 (0x0800)
    [Stream index: 2]
Internet Protocol Version 4, Src: 192.168.12.2, Dst: 224.0.0.5
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
        1100 00.. = Differentiated Services Codepoint: Class
Selector 6 (48)
        .... ..00 = Explicit Congestion Notification: Not ECN-
Capable Transport (0)
    Total Length: 76
    Identification: 0x027d (637)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 1
    Protocol: OSPF IGP (89)
    Header Checksum: 0x096d [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.12.2
    Destination Address: 224.0.0.5
    [Stream index: 0]
Open Shortest Path First
    OSPF Header
        Version: 2
        Message Type: LS Update (4) — тип
        Packet Length: 56
        Source OSPF Router: 2.2.2.2
        Area ID: 0.0.0.0 (Backbone)
        Checksum: 0x4a6d [correct]
        Instance ID: Base IPv4 Unicast Instance (0)
        Auth Type: Null (0)
        Auth Data (none): 0000000000000000
    LS Update Packet
        Number of LSAs: 1
        LSA-type 3 (Summary-LSA (IP network)), len 28
            .000 0000 0000 0001 = LS Age (seconds): 1
            0... .... .... .... = Do Not Age Flag: 0
            Options: 0x22, (DC) Demand Circuits, (E) External
Routing
                0... .... = DN: Not set
                .0.. .... = (O) Opaque: Not set
                ..1. .... = (DC) Demand Circuits: Supported — может
```

циклические сети поддреживать(?)

```
...0 .... = (L) LLS Data block: Not Present
.... 0... = (N) NSSA: Not supported
.... .0.. = (MC) Multicast: Not capable
.... ..1. = (E) External Routing: Capable
```
— может маршрутизировать извне

```
.... ...0 = (MT) Multi-Topology Routing: No
LS Type: Summary-LSA (IP network) (3)
Link State ID: 192.168.11.
```
0 – сеть которая это может

```
Advertising Router: 2.2.2.2
Sequence Number: 0x80000002
Checksum: 0x3e7e
Length: 28
Netmask: 255.255.255.0
TOS: 0
Metric: 3
```

LS Acknowledge: это сообщение от маршрутизатора о том, что он записал предложенную информацию

```
Frame 145: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits)
on interface -, id 0
Ethernet II, Src: cc:01:63:03:00:10 (cc:01:63:03:00:10), Dst: IPv4mcast_05
(01:00:5e:00:00:05)
    Destination: IPv4mcast_05 (01:00:5e:00:00:05)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: cc:01:63:03:00:10 (cc:01:63:03:00:10)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 3]
Internet Protocol Version 4, Src: 192.168.12.1, Dst: 224.0.0.5
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
        1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable
Transport (0)
    Total Length: 84
    Identification: 0x0190 (400)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 1
    Protocol: OSPF IGP (89)
    Header Checksum: 0x0a53 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.12.1
    Destination Address: 224.0.0.5
    [Stream index: 1]
Open Shortest Path First
    OSPF Header
        Version: 2
```

```
        Message Type: LS Acknowledge (5)
```
- тип

```
        Packet Length: 64
        Source OSPF Router: 1.1.1.1
        Area ID: 0.0.0.0 (Backbone)
        Checksum: 0x7d3b [correct]
        Instance ID: Base IPv4 Unicast Instance (0)
        Auth Type: Null (0)
        Auth Data (none): 0000000000000000
    LSA-type 3 (Summary-LSA (IP network)), len 28
```
– пакет с принятыми предложениями

```
        .000 0000 0000 0001 = LS Age (seconds): 1
        0... .... .... .... = Do Not Age Flag: 0
        Options: 0x22, (DC) Demand Circuits, (E) External Routing
            0... .... = DN: Not set
            .0.. .... = (O) Opaque: Not set
            ..1. .... = (DC) Demand Circuits: Supported
            ...0 .... = (L) LLS Data block: Not Present
            .... 0... = (N) NSSA: Not supported
            .... .0.. = (MC) Multicast: Not capable
            .... ..1. = (E) External Routing: Capable
            .... ...0 = (MT) Multi-Topology Routing: No
        LS Type: Summary-LSA (IP network) (3)
        Link State ID: 192.168.11.0
        Advertising Router: 2.2.2.2
        Sequence Number: 0x80000002
        Checksum: 0x3e7e
        Length: 28
    LSA-type 3 (Summary-LSA (IP network)), len 28
```
– пакет с принятыми предложениями

```
        .000 0000 0000 0001 = LS Age (seconds): 1
        0... .... .... .... = Do Not Age Flag: 0
        Options: 0x22, (DC) Demand Circuits, (E) External Routing
            0... .... = DN: Not set
            .0.. .... = (O) Opaque: Not set
            ..1. .... = (DC) Demand Circuits: Supported
            ...0 .... = (L) LLS Data block: Not Present
            .... 0... = (N) NSSA: Not supported
            .... .0.. = (MC) Multicast: Not capable
            .... ..1. = (E) External Routing: Capable
            .... ...0 = (MT) Multi-Topology Routing: No
        LS Type: Summary-LSA (IP network) (3)
        Link State ID: 192.168.8.0
        Advertising Router: 2.2.2.2
        Sequence Number: 0x80000002
        Checksum: 0x5f60
        Length: 28
```

7) Сохранить в отдельные файлы с префиксом rt_ и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.